Prompts Are Design Instructions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Prompts Are Design Instructions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Prompts Are Design Instructions trace and record the first step you can explain without help.

Use W2: Score a response by separate criteria for Specifying and testing **prompt**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Prompts Are Design Instructions explanation against the supplied facts before discussing it.

Use the worked case for Specifying and testing **prompt**s. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Specify and test a new exhibit card

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Prompts Are Design Instructions target to a trace and a justified conclusion.

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.

What should happen if the paragraph does not explain evaporation?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Prompts Are Design Instructions case and show the deciding evidence.

The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

Write a checkable **prompt**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Improve this **prompt**: tell me about space. Specify a reader, a focused task and an **output** format suitable for a young learner.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Prompts Are Design Instructions adds to Helpful Classroom Sorter and keep the evidence.

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.

What should happen if the paragraph does not explain evaporation?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Find what the vague request leaves open

Find what the vague request leaves open. Apply the supplied Prompts Are Design Instructions facts and keep your reasoning.

The request says “Write something good about our club.” Name four decisions it leaves open. Then explain how source C, the Grade 6 **audience** and four labelled lines make the task easier to check.

My reasoning and evidence

W2 Score a response by separate criteria

> Score a response by separate criteria. Apply the supplied Prompts Are Design Instructions facts and keep your reasoning.

For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and **evidence** checks pass or fail. Is the card acceptable overall? Explain.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Handle disagreement without choosing a favourite

> Handle disagreement without choosing a favourite. Apply the supplied Prompts Are Design Instructions facts and keep your reasoning.

C gives Room 7 and D gives Room 9 for the same club event, with no replacement or priority stated. Both agree on Monday and paper, and neither gives a time.

Under P2, write the four **output** values and explain why the room cannot be selected from confidence or wording alone.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Specify and test a new exhibit card

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Specify and test a new exhibit card. Apply the supplied Prompts Are Design Instructions facts and keep your reasoning.

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Preserve the comparison evidence

> Preserve the comparison evidence. Apply the supplied Prompts Are Design Instructions facts and keep your reasoning.

A pupil tries P1 once and P2 five times, then compares the P1 response only with the best P2 response and declares P2 always reliable. Identify two problems and design a fairer small comparison.

State a conclusion the **evidence** could support.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Prompts Are Design Instructions

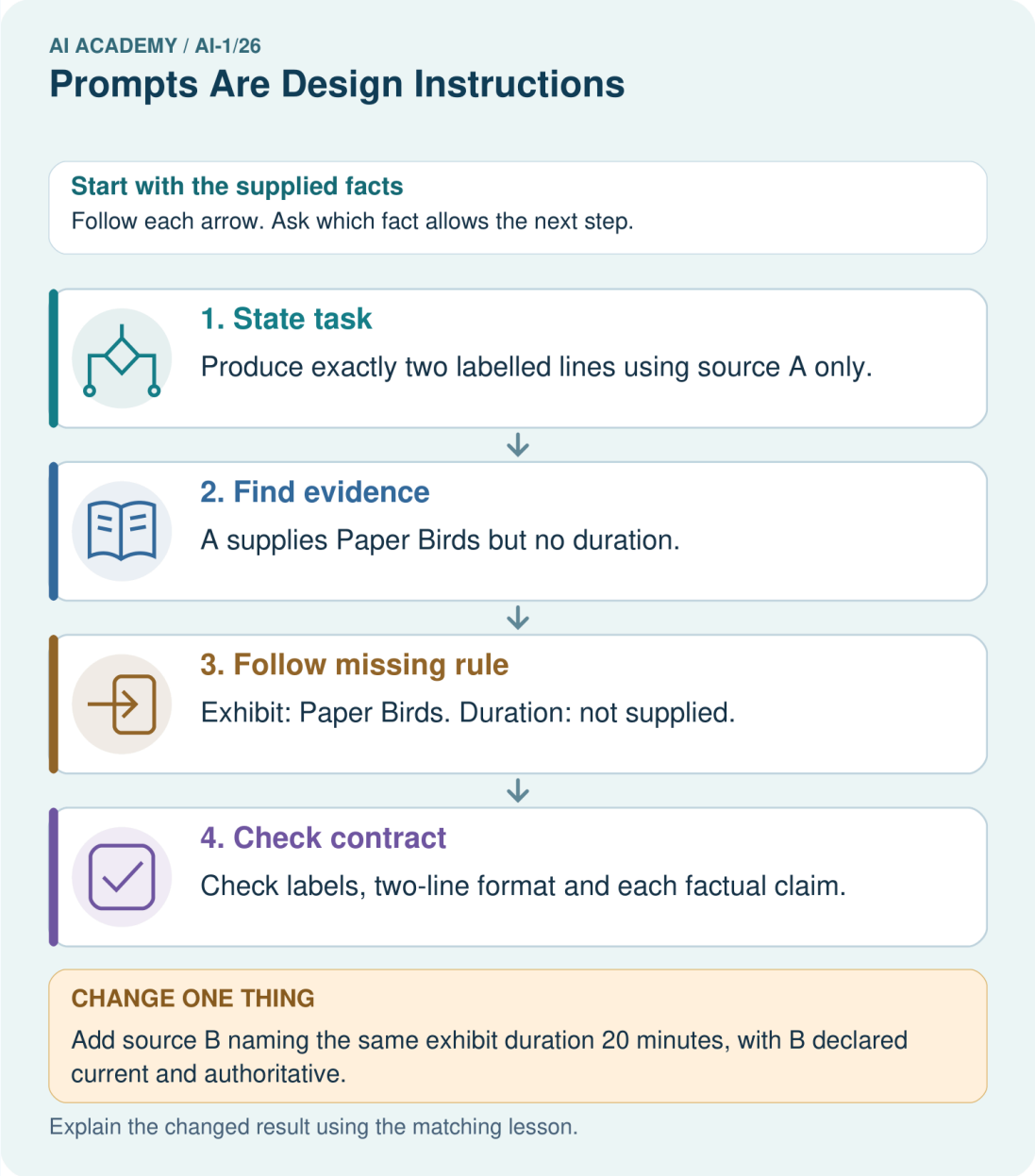
> A fictional club needs a short exhibit card, without guesses. Explain the labelled step before

> checking the changed condition.

A fictional club needs a short exhibit card, without guesses.

Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.



Follow the labelled picture with the matching explanation. Open the vector diagram

What must happen if B instead conflicts with another equally authoritative duration record?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional club needs a short exhibit card, without guesses.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check labels, two-line format and each factual claim.

1. State task

2. Follow missing rule

3. Find **evidence**

4. Check contract

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check contract Check labels, two-line format and each factual claim. A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient.

Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts. Check content support separately from whether the answer looks tidy.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Duration becomes 20 minutes.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Duration becomes 20 minutes. There is now an authorised source for the previously missing field.

Repair a missing building block

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

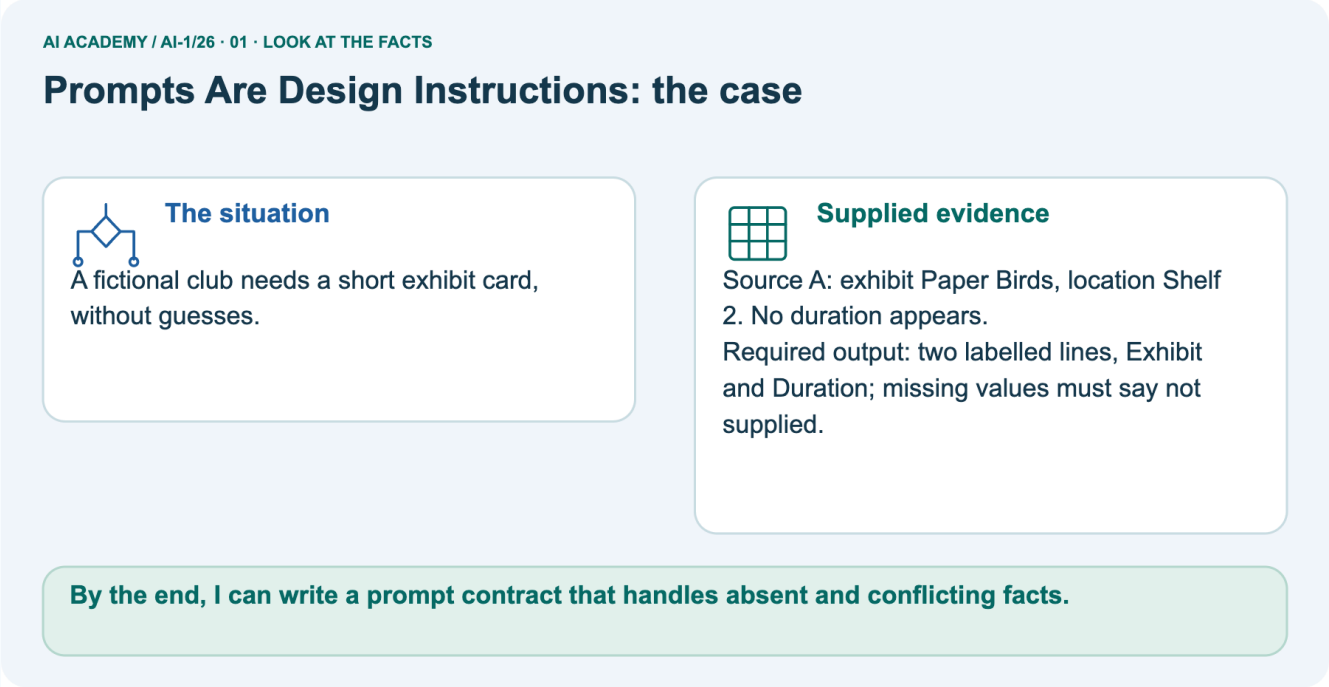
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Prompts Are Design Instructions picture from its labels, then record a relevant limit.

01. Prompts Are Design Instructions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

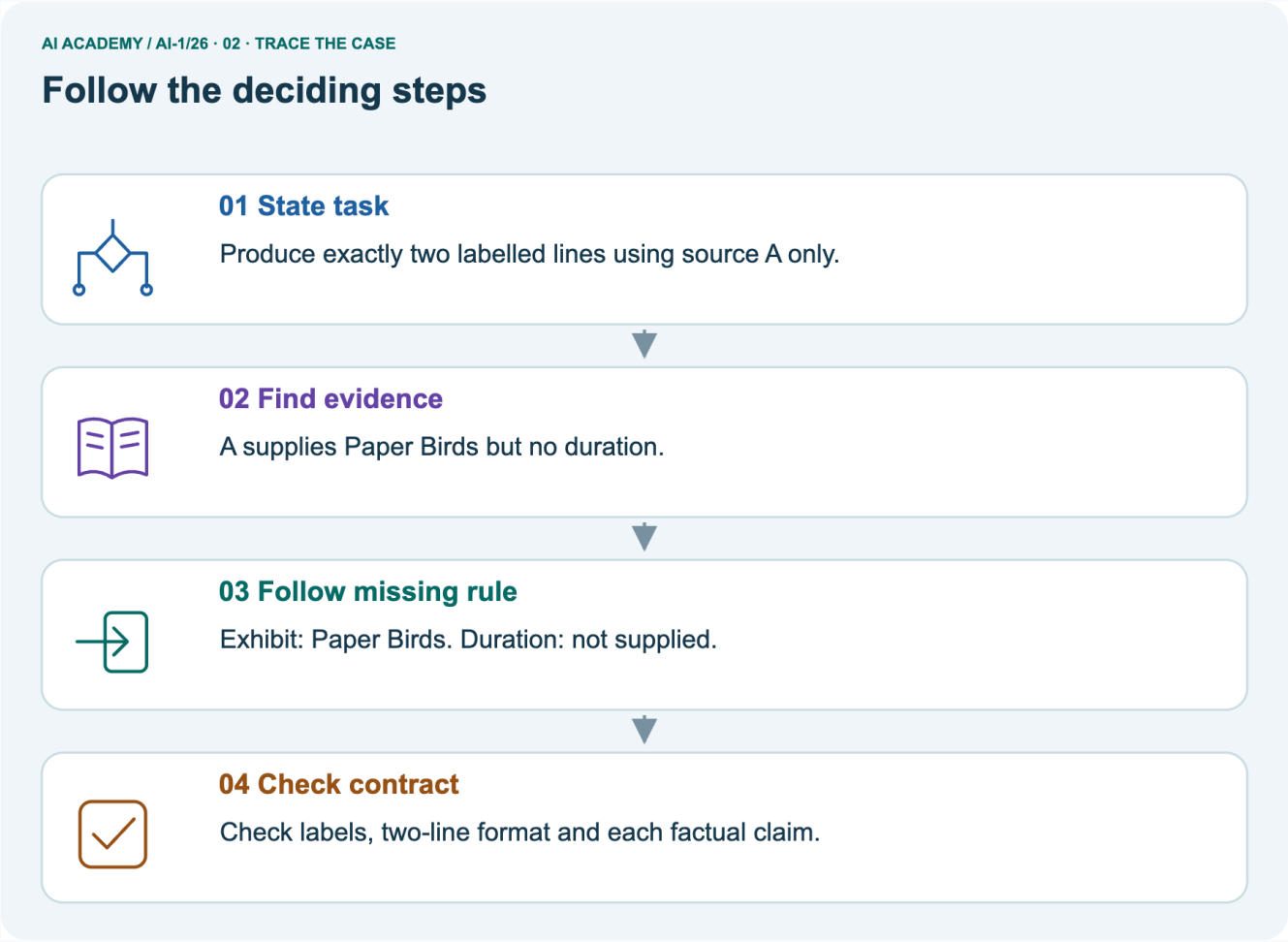
A fictional club needs a short exhibit card, without guesses. Supplied facts: Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State task: Produce exactly two labelled lines using source A only. → Find **evidence**: A supplies Paper Birds but no duration. → Follow missing rule: Exhibit: Paper Birds.

Duration: not supplied. → Check contract: Check labels, two-line format and each factual claim.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. From prompt requirement to evidence of success

AI ACADEMY / AI-1/26 · 03 · INSPECT THE LESSON MODEL

From prompt requirement to evidence of success

Requirement	Expected behaviour	How to check
Use source C only	Monday and Room 7	Compare with C
Mark missing values	Time: not stated	Check absent time
Four labelled lines	Day Time Place Bring	Count and inspect
No added details	No invented promises	Check every claim

Choose these checks before reading a response. A response must satisfy all required criteria, not merely look tidy.

Follow the labelled picture with the matching explanation. Open the vector diagram

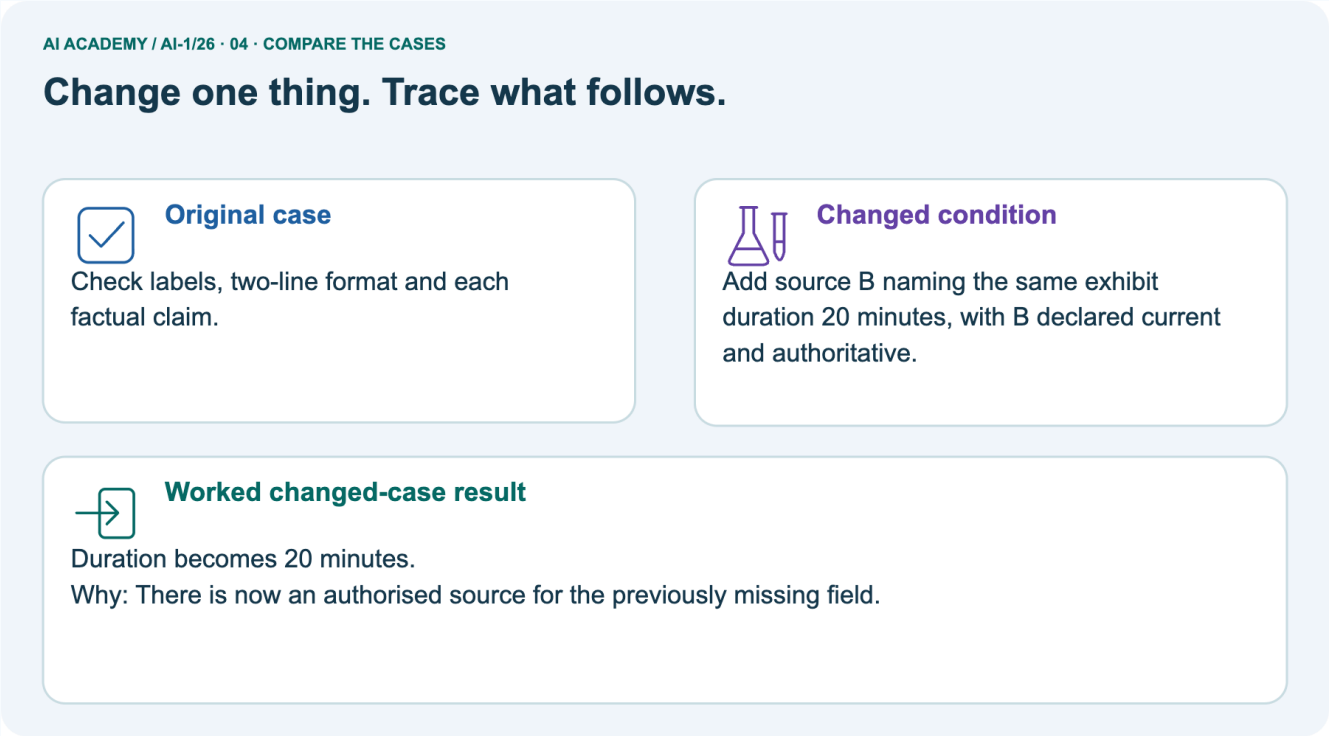
From **prompt** requirement to **evidence** of success. Use source C only | Monday and Room 7 | Compare with C. Mark missing values | Time: not stated | Check absent time.

Four labelled lines | Day Time Place Bring | Count and inspect. No added details | No invented promises | Check every claim

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

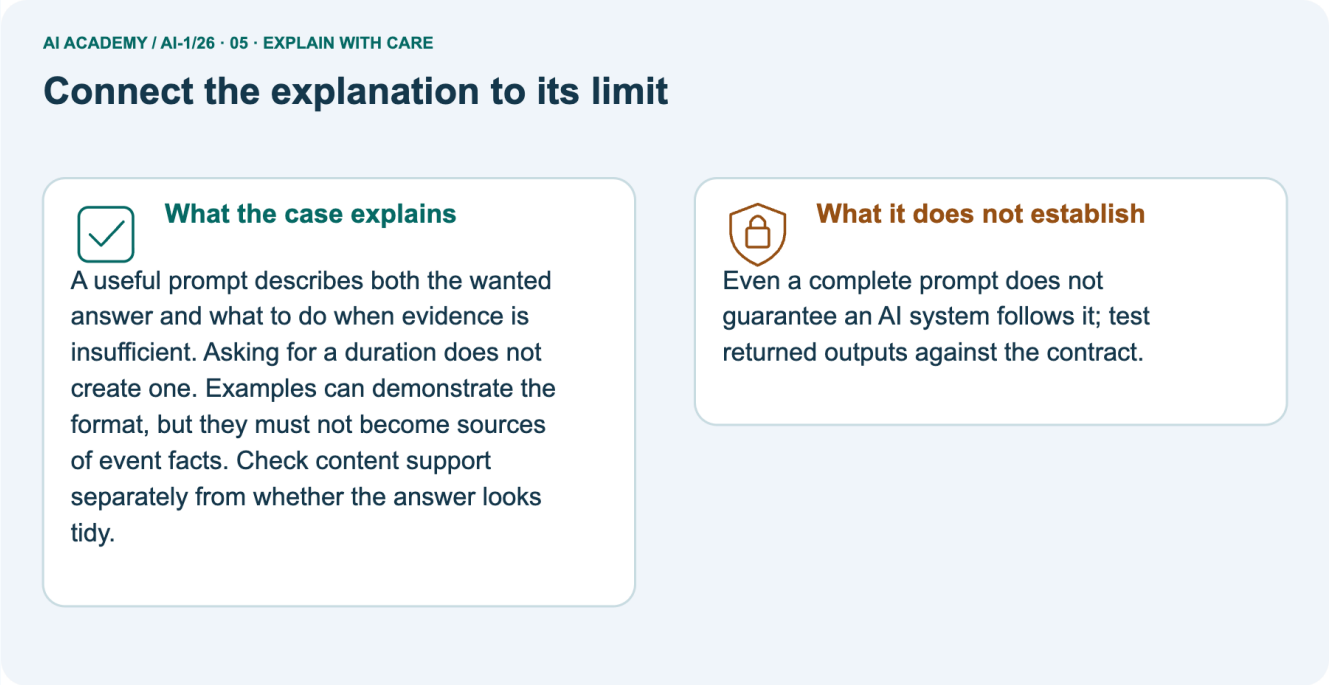
Original case: Check labels, two-line format and each factual claim. Changed condition: Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Expected result: Duration becomes 20 minutes. Reason: There is now an authorised source for the previously missing field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

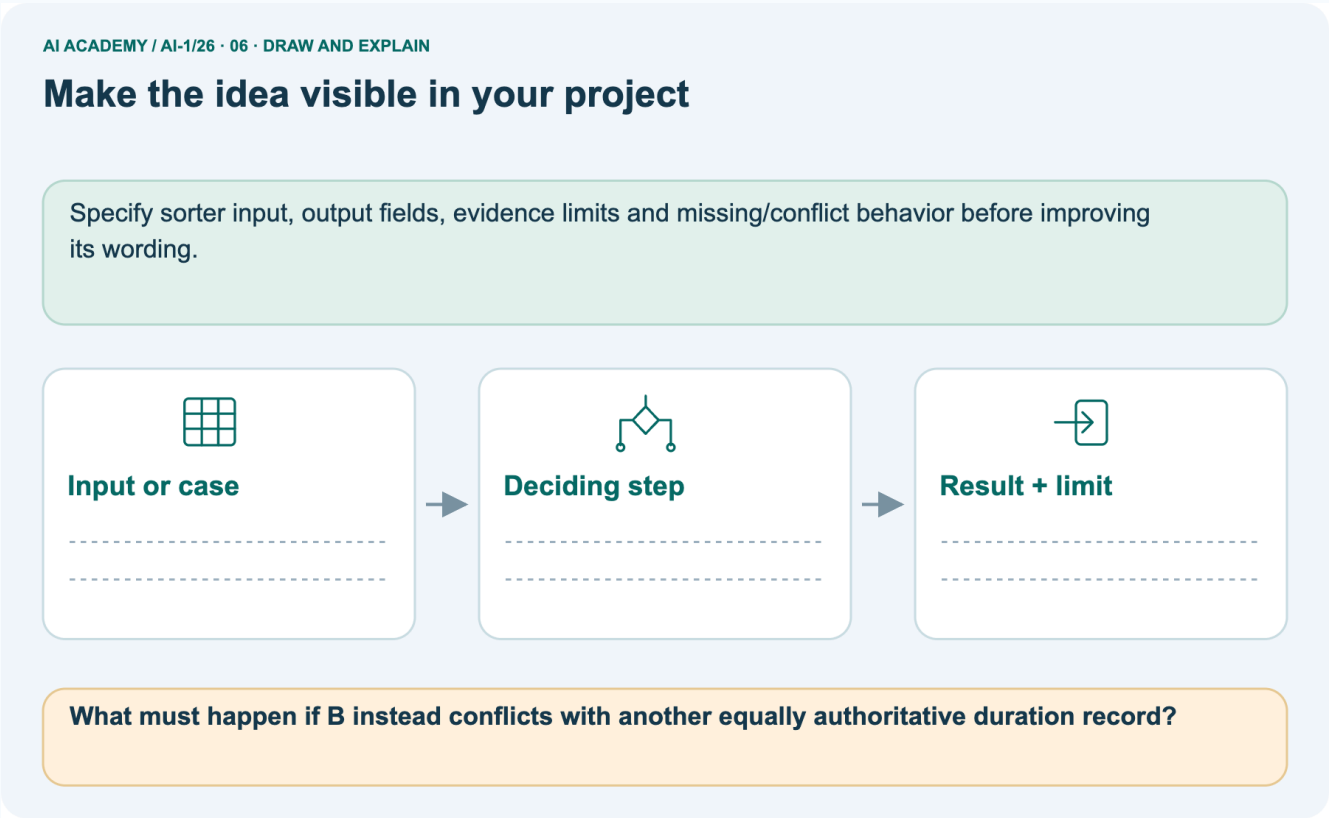
A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy. Limit: Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording. Draw your **input**, deciding step, result and limit.

Supported question: What must happen if B instead conflicts with another equally authoritative duration record?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.